Problem 1 (§12.1, #6 (adapted))**.** Suppose $f : \mathbb{Z} \to \mathbb{Z}$ is defined as $f = \{(x, 4x+5) : x \in \mathbb{Z}\}$.

a) State the domain, codomain and range of f.

b) Find $f(X)$ where $X = \{2k : k \in \mathbb{Z}\}$.

c) Find $f^{-1}(Y)$ where $Y = \{2k : k \in \mathbb{Z}\}$.

Problem 2 (§12.1, #10)**.** Consider the set $f = \left\{(x^3, x) : x \in \mathbb{R}\right\}$. Is this a function from $\mathbb{R}$ to $\mathbb{R}$? Explain.

Problem 3 (§12.2, #8)**.** A function $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z} \times \mathbb{Z}$ is defined as $f(m, n) = (m + n, 2m + n)$. Verify

a) whether this function is injective, and

b) whether it is surjective.

Problem 4 (§12.2, #14, §12.5 #8)**.** Consider the function $\theta : \mathcal{P}(\mathbb{Z}) \to \mathcal{P}(\mathbb{Z})$ defined as $\theta(X) = \overline{X}$.

a) Is θ injective?

b) Is θ surjective?

c) Is θ bijective? If so, find θ^{-1}.

Problem 5 (§12.4, #8)**.** Consider the functions $f, g : \mathbb{Z}\times\mathbb{Z} \to \mathbb{Z}\times\mathbb{Z}$ defined as $f(m, n) = (3m-4n, 2m+n)$ and $g(m, n) = (5m + n, m)$. Find the formulas for

a) $g \circ f$

b) $f \circ g$

c) $f \circ f$

d) $g \circ g$

Problem 6 (§12.5, #6)**.** The function $f : \mathbb{Z}\times\mathbb{Z} \to \mathbb{Z}\times\mathbb{Z}$ defined by the formula $f(m, n) = (5m+4n, 4m+3n)$ is bijective. Find its inverse.

Problem 7 (§12.6, #10)**.** Given $f : A \to B$ and subsets $Y, Z \subseteq B$. Prove that

$$f^{-1}(Y \cap Z) = f^{-1}(Y) \cap f^{-1}(Z).$$

Problem 8 (§12.6, #12)**.** Consider $f : A \to B$. Prove that

a) f is injective if and only if $X = f^{-1}(f(X))$ for all $X \subseteq A$.

b) f is surjective if and only if $f(f^{-1}(Y)) = Y$ for all $Y \subseteq B$.